

Backpropagation Through Time (BPTT)

Single Epoch for the 5-Time-Step Example — Two Stacked LSTM Units (Anomaly Detection Network)

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This continues directly from the two-unit forward-pass document. Part I below is the complete backpropagation of Unit 1, unchanged from the original document. Part II adds the complete backpropagation of Unit 2 (the top unit of the anomaly detection stack) for all 5 time steps, with every intermediate calculation shown.

Part I: Backpropagation of Unit 1 (Unchanged from the Original Document)

Loss Definition (on final output only)

We define a simple loss on the final hidden state h_5 :

$$L = \frac{1}{2}(h_5 - y)^2$$

where the target $y = 0.25$ (chosen as a reasonable value near the forward-pass result $h_5 \approx 0.2246$).

Gradient w.r.t. final output:

$$\frac{\partial L}{\partial h_5} = h_5 - 0.25 \approx 0.2246 - 0.25 = -0.0254$$

We now propagate this gradient backwards through time from $t = 5$ to $t = 1$, using the derivatives from the previous document. Gradients for the 12 shared weights are accumulated across all time steps.

Common Building Blocks (from previous document)

For every time step we first compute these (using forward-pass values):

- $\frac{\partial h_t}{\partial o_t} = \tanh(C_t)$
- $\frac{\partial h_t}{\partial C_t} = o_t \odot (1 - \tanh^2(C_t))$
- $\frac{\partial h_t}{\partial f_t} = \frac{\partial h_t}{\partial C_t} \odot C_{t-1}$
- $\frac{\partial h_t}{\partial i_t} = \frac{\partial h_t}{\partial C_t} \odot \tilde{C}_t$
- $\frac{\partial h_t}{\partial \tilde{C}_t} = \frac{\partial h_t}{\partial C_t} \odot i_t$

(where $\sigma'(z) = \sigma(z) \odot (1 - \sigma(z))$ and $\tanh'(z) = 1 - \tanh^2(z)$).

Backpropagation from $t = 5$ (starting point)

Using forward-pass values at $t = 5$:

$h_5 \approx 0.2246$, $C_5 \approx 0.3814$, $o_5 \approx 0.6169$, $f_5 \approx 0.5053$, $i_5 \approx 0.6244$, $\tilde{C}_5 \approx 0.3130$, $h_4 \approx 0.2184$, $C_4 \approx 0.3681$, $x_5 = 0.6$

Common building blocks at $t = 5$:

- $\frac{\partial h_5}{\partial o_5} = \tanh(0.3814) \approx 0.3640$
- $\frac{\partial h_5}{\partial C_5} = 0.6169 \odot (1 - 0.3640^2) \approx 0.5352$
- $\frac{\partial h_5}{\partial f_5} = 0.5352 \odot 0.3681 \approx 0.1970$
- $\frac{\partial h_5}{\partial i_5} = 0.5352 \odot 0.3130 \approx 0.1675$
- $\frac{\partial h_5}{\partial \tilde{C}_5} = 0.5352 \odot 0.6244 \approx 0.3342$

Output Gate Gradients

Pre-activation at $t = 5$: $w_{ho} \cdot h_4 + w_{xo} \cdot x_5 + b_o \approx 0.4766$

$\sigma'(0.4766) \approx 0.2363$

- $\frac{\partial h_5}{\partial w_{ho}} = (0.3640 \odot 0.2363) \times 0.2184 \approx 0.0188$
- $\frac{\partial L}{\partial w_{ho}}$ (from $t = 5$) ≈ -0.000478
- $\frac{\partial h_5}{\partial w_{xo}} = (0.3640 \odot 0.2363) \times 0.6 \approx 0.0516$
- $\frac{\partial L}{\partial w_{xo}}$ (from $t = 5$) ≈ -0.001311
- $\frac{\partial h_5}{\partial b_o} = 0.3640 \odot 0.2363 \approx 0.0860$
- $\frac{\partial L}{\partial b_o}$ (from $t = 5$) ≈ -0.002184

Forget Gate Gradients

Pre-activation at $t = 5$: $w_{hf} \cdot h_4 + w_{xf} \cdot x_5 + b_f \approx 0.0212$

$\sigma'(0.0212) \approx 0.2500$

- $\frac{\partial h_5}{\partial w_{hf}} = (0.1970 \odot 0.2500) \times 0.2184 \approx 0.01076$
- $\frac{\partial L}{\partial w_{hf}}$ (from $t = 5$) ≈ -0.000273

- $\frac{\partial h_5}{\partial w_{xf}} = (0.1970 \odot 0.2500) \times 0.6 \approx 0.02955$
- $\frac{\partial L}{\partial w_{xf}}$ (from $t = 5$) ≈ -0.000750
- $\frac{\partial h_5}{\partial b_f} = 0.1970 \odot 0.2500 \approx 0.04925$
- $\frac{\partial L}{\partial b_f}$ (from $t = 5$) ≈ -0.001251

Input-Gate Sigmoid Gradients

Pre-activation: $w_{hi1} \cdot h_4 + w_{xi1} \cdot x_5 + b_{i1} \approx 0.5084$

$\sigma'(0.5084) \approx 0.2347$

- $\frac{\partial h_5}{\partial w_{hi1}} = (0.1675 \odot 0.2347) \times 0.2184 \approx 0.00859$
- $\frac{\partial L}{\partial w_{hi1}}$ (from $t = 5$) ≈ -0.000218
- $\frac{\partial h_5}{\partial w_{xi1}} = (0.1675 \odot 0.2347) \times 0.6 \approx 0.02358$
- $\frac{\partial L}{\partial w_{xi1}}$ (from $t = 5$) ≈ -0.000599
- $\frac{\partial h_5}{\partial b_{i1}} = 0.1675 \odot 0.2347 \approx 0.0393$
- $\frac{\partial L}{\partial b_{i1}}$ (from $t = 5$) ≈ -0.0010

Candidate Gradients

Pre-activation: $w_{hi2} \cdot h_4 + w_{xi2} \cdot x_5 + b_{i2} \approx 0.3240$

$\tanh'(0.3240) \approx 0.902$

- $\frac{\partial h_5}{\partial w_{hi2}} = (0.3342 \odot 0.902) \times 0.2184 \approx 0.0658$
- $\frac{\partial L}{\partial w_{hi2}}$ (from $t = 5$) ≈ -0.00167
- $\frac{\partial h_5}{\partial w_{xi2}} = (0.3342 \odot 0.902) \times 0.6 \approx 0.1808$
- $\frac{\partial L}{\partial w_{xi2}}$ (from $t = 5$) ≈ -0.00459
- $\frac{\partial h_5}{\partial b_{i2}} = 0.3342 \odot 0.902 \approx 0.3014$
- $\frac{\partial L}{\partial b_{i2}}$ (from $t = 5$) ≈ -0.00766

Recurrent Gradients at $t = 5$ (to pass to $t = 4$)

$$\begin{aligned}\frac{\partial h_5}{\partial h_4} &= \left(\frac{\partial h_5}{\partial o_5} \odot \sigma'(\text{net}_{o5}) \right) \cdot w_{ho} \\ &+ \left(\frac{\partial h_5}{\partial f_5} \odot \sigma'(\text{net}_{f5}) \right) \cdot w_{hf} \\ &+ \left(\frac{\partial h_5}{\partial i_5} \odot \sigma'(\text{net}_{i5}) \right) \cdot w_{hi1} \\ &+ \left(\frac{\partial h_5}{\partial \tilde{C}_5} \odot \tanh'(\text{net}_{\tilde{C}_5}) \right) \cdot w_{hi2}\end{aligned}$$

(Computed numerically ≈ 0.312)

$$\frac{\partial h_5}{\partial C_4} = \frac{\partial h_5}{\partial C_5} \odot f_5 \approx 0.2704$$

Then:

$$\begin{aligned}\frac{\partial L}{\partial h_4} &= (-0.0254) \times 0.312 \approx -0.00792 \\ \frac{\partial L}{\partial C_4} &= (-0.0254) \times 0.2704 \approx -0.00687\end{aligned}$$

Propagation to $t = 4, 3, 2, 1$ (Same Process)

Repeat the exact same procedure for each earlier time step:

1. Use the incoming $\frac{\partial L}{\partial h_t}$ and $\frac{\partial L}{\partial C_t}$ (from the next time step).
2. Compute the five common building blocks at that t .
3. Compute the 12 local weight gradients using the four gate sections.
4. Accumulate them to the running total for each weight.
5. Compute the new $\frac{\partial L}{\partial h_{t-1}}$ and $\frac{\partial L}{\partial C_{t-1}}$ using the recurrent formulas, and pass them backwards.

All 12 weights receive accumulated gradients from every time step where they had an effect.

Final Accumulated Gradients (after processing $t = 5 \rightarrow t = 1$)

- $\frac{\partial L}{\partial w_{hf}} \approx -0.0012$
- $\frac{\partial L}{\partial w_{xf}} \approx -0.0031$
- $\frac{\partial L}{\partial b_f} \approx -0.0048$
- $\frac{\partial L}{\partial w_{hi1}} \approx -0.0009$
- $\frac{\partial L}{\partial w_{xi1}} \approx -0.0025$

- $\frac{\partial L}{\partial b_{i1}} \approx -0.0041$
- $\frac{\partial L}{\partial w_{hi2}} \approx -0.0068$
- $\frac{\partial L}{\partial w_{xi2}} \approx -0.0182$
- $\frac{\partial L}{\partial b_{i2}} \approx -0.0295$
- $\frac{\partial L}{\partial w_{ho}} \approx -0.0019$
- $\frac{\partial L}{\partial w_{xo}} \approx -0.0053$
- $\frac{\partial L}{\partial b_o} \approx -0.0087$

Weight Update (Gradient Descent, Single Epoch)

Using learning rate $\alpha = 0.1$:

For each weight:

$$w_{\text{new}} = w_{\text{old}} - \alpha \times \frac{\partial L}{\partial w}$$

Example updates:

- $w_{hf} \leftarrow 0.28 - 0.1 \times (-0.0012) \approx 0.28012$
- $w_{xf} \leftarrow -0.15 - 0.1 \times (-0.0031) \approx -0.14969$
- $b_f \leftarrow 0.05 - 0.1 \times (-0.0048) \approx 0.05048$

(and similarly for all other weights).

Part II: Backpropagation of Unit 2 (Top of the Anomaly Detection Stack)

Unit 2's forward pass was computed in the two-unit forward-pass document: its input at each time step is Unit 1's hidden state $h_t^{(1)}$, and its final output is $h_5^{(2)} \approx 0.1325$.

Loss Definition for the Anomaly Detector (on final output only)

The network output $h_5^{(2)}$ is compared against the expected “normal” value y_2 ; the squared prediction error is both the anomaly score and the training loss:

$$L_2 = \frac{1}{2} \left(h_5^{(2)} - y_2 \right)^2$$

where the target $y_2 = 0.10$ (chosen as a reasonable value near the forward-pass result $h_5^{(2)} \approx 0.1325$). Training on normal sequences drives this score toward zero, so that at detection time a large score flags an anomaly.

Gradient w.r.t. final output:

$$\frac{\partial L_2}{\partial h_5^{(2)}} = h_5^{(2)} - 0.10 \approx 0.1325 - 0.10 = 0.0325$$

We propagate this gradient backwards through time from $t = 5$ to $t = 1$ inside Unit 2, exactly as in Part I. Gradients for Unit 2's 12 shared weights are accumulated across all time steps.

Common Building Blocks (same as Part I, applied to Unit 2)

The five building blocks and the derivatives $\sigma'(z) = \sigma(z) \odot (1 - \sigma(z))$ and $\tanh'(z) = 1 - \tanh^2(z)$ are identical; only the values come from Unit 2's forward pass, and the input x_t is replaced everywhere by $h_t^{(1)}$.

At every time step below, the 12 weight gradients are written using the four pre-activation gradients (“deltas”), which combine a building block with the matching activation derivative and, for the f , i , and $\tilde{C}$ gates, the total cell-state gradient:

$$\begin{aligned} \delta_{ot} &= \frac{\partial L_2}{\partial h_t^{(2)}} \odot \tanh(C_t^{(2)}) \odot \sigma'(\text{net}_{ot}) & \delta_{ft} &= \frac{\partial L_2}{\partial C_t^{(2)}} \odot C_{t-1}^{(2)} \odot \sigma'(\text{net}_{ft}) \\ \delta_{it} &= \frac{\partial L_2}{\partial C_t^{(2)}} \odot \tilde{C}_t^{(2)} \odot \sigma'(\text{net}_{it}) & \delta_{\tilde{C}t} &= \frac{\partial L_2}{\partial C_t^{(2)}} \odot i_t^{(2)} \odot \tanh'(\text{net}_{\tilde{C}t}) \end{aligned}$$

Each weight gradient is then the delta multiplied by the input that weight touches ($h_{t-1}^{(2)}$ for the w_h weights, $h_t^{(1)}$ for the w_x weights, and 1 for the biases). This is the same computation as the gate sections of Part I, written so that the incoming $\frac{\partial L_2}{\partial C_t^{(2)}}$ channel from the next time step is included at $t < 5$.

Full Backpropagation of Unit 2 with Every Intermediate Step

Time Step $t = 5$

Using Unit 2 forward-pass values at $t = 5$: $h_5^{(2)} \approx 0.1325$, $C_5^{(2)} \approx 0.2334$, $o_5^{(2)} \approx 0.5778$, $f_5^{(2)} \approx 0.5158$, $i_5^{(2)} \approx 0.5599$, $\tilde{C}_5^{(2)} \approx 0.2190$, $h_4^{(2)} \approx 0.1218$, $C_4^{(2)} \approx 0.2148$, input $h_5^{(1)} \approx 0.2246$

Incoming gradients (from the next time step)

- $\frac{\partial L_2}{\partial h_5^{(2)}} \approx 0.032465$
- $\frac{\partial L_2}{\partial C_5^{(2)}} \Big|_{\text{incoming}} \approx 0.000000$

Common building blocks

- $\tanh(C_5^{(2)}) = \tanh(0.2334) \approx 0.2292, \quad 1 - \tanh^2(C_5^{(2)}) \approx 0.9474$
- $\frac{\partial h_5}{\partial o_5} = \tanh(C_5^{(2)}) \approx 0.2292$
- $\frac{\partial h_5}{\partial C_5} = 0.5778 \odot 0.9474 \approx 0.5475$
- $\frac{\partial h_5}{\partial f_5} = 0.5475 \odot 0.2148 \approx 0.1176$
- $\frac{\partial h_5}{\partial i_5} = 0.5475 \odot 0.2190 \approx 0.1199$
- $\frac{\partial h_5}{\partial \tilde{C}_5} = 0.5475 \odot 0.5599 \approx 0.3065$

Total cell-state gradient (both channels combined)

$$\frac{\partial L_2}{\partial C_5^{(2)}} = 0.032465 \times 0.5475 + 0.000000 \approx 0.017773$$

Activation derivatives at the pre-activations

- $\sigma'(\text{net}_{f5} = 0.0630) \approx 0.2498$
- $\sigma'(\text{net}_{i5} = 0.2407) \approx 0.2464$
- $\sigma'(\text{net}_{o5} = 0.3138) \approx 0.2439$
- $\tanh'(\text{net}_{\tilde{C}5} = 0.2226) \approx 0.9521$

Pre-activation gradients (the four deltas)

- $\delta_{o5} = 0.032465 \times 0.2292 \times 0.2439 \approx 0.001816$
- $\delta_{f5} = 0.017773 \times 0.2148 \times 0.2498 \approx 0.000954$
- $\delta_{i5} = 0.017773 \times 0.2190 \times 0.2464 \approx 0.000959$
- $\delta_{\tilde{C}5} = 0.017773 \times 0.5599 \times 0.9521 \approx 0.009474$

Output Gate Gradients

- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} (\text{from } t = 5) = 0.001816 \times 0.1218 \approx 0.000221$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} (\text{from } t = 5) = 0.001816 \times 0.2246 \approx 0.000408$

- $\frac{\partial L_2}{\partial b_o^{(2)}} \text{ (from } t = 5) = 0.001816 \times 1 \approx 0.001816$

Forget Gate Gradients

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \text{ (from } t = 5) = 0.000954 \times 0.1218 \approx 0.000116$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \text{ (from } t = 5) = 0.000954 \times 0.2246 \approx 0.000214$
- $\frac{\partial L_2}{\partial b_f^{(2)}} \text{ (from } t = 5) = 0.000954 \times 1 \approx 0.000954$

Input-Gate Sigmoid Gradients

- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \text{ (from } t = 5) = 0.000959 \times 0.1218 \approx 0.000117$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \text{ (from } t = 5) = 0.000959 \times 0.2246 \approx 0.000215$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \text{ (from } t = 5) = 0.000959 \times 1 \approx 0.000959$

Candidate Gradients

- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \text{ (from } t = 5) = 0.009474 \times 0.1218 \approx 0.001154$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \text{ (from } t = 5) = 0.009474 \times 0.2246 \approx 0.002128$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \text{ (from } t = 5) = 0.009474 \times 1 \approx 0.009474$

Recurrent Gradients at $t = 5$ (to pass to $t = 4$)

$$\frac{\partial L_2}{\partial h_4^{(2)}} = 0.001816 \times 0.41 + 0.000954 \times 0.33 + 0.000959 \times 0.29 + 0.009474 \times 0.18 \approx 0.003042$$

$$\frac{\partial L_2}{\partial C_4^{(2)}} = 0.017773 \times 0.5158 \approx 0.009167$$

Gradient passed down to Unit 1 at $t = 5$ (through the input connection $h_5^{(1)}$):

$$\frac{\partial L_2}{\partial h_5^{(1)}} = 0.001816 \times 0.24 + 0.000954 \times (-0.21) + 0.000959 \times 0.38 + 0.009474 \times 0.27 \approx 0.003158$$

Time Step $t = 4$

Using Unit 2 forward-pass values at $t = 4$: $h_4^{(2)} \approx 0.1218$, $C_4^{(2)} \approx 0.2148$, $o_4^{(2)} \approx 0.5757$, $f_4^{(2)} \approx 0.5146$, $i_4^{(2)} \approx 0.5580$, $\tilde{C}_4^{(2)} \approx 0.2144$, $h_3^{(2)} \approx 0.1043$, $C_3^{(2)} \approx 0.1850$, input $h_4^{(1)} \approx 0.2184$

Incoming gradients (from the next time step)

- $\frac{\partial L_2}{\partial h_4^{(2)}} \approx 0.003042$
- $\frac{\partial L_2}{\partial C_4^{(2)}} \Big|_{\text{incoming}} \approx 0.009167$

Common building blocks

- $\tanh(C_4^{(2)}) = \tanh(0.2148) \approx 0.2116, \quad 1 - \tanh^2(C_4^{(2)}) \approx 0.9552$
- $\frac{\partial h_4}{\partial o_4} = \tanh(C_4^{(2)}) \approx 0.2116$
- $\frac{\partial h_4}{\partial C_4} = 0.5757 \odot 0.9552 \approx 0.5499$
- $\frac{\partial h_4}{\partial f_4} = 0.5499 \odot 0.1850 \approx 0.1017$
- $\frac{\partial h_4}{\partial i_4} = 0.5499 \odot 0.2144 \approx 0.1179$
- $\frac{\partial h_4}{\partial \tilde{C}_4} = 0.5499 \odot 0.5580 \approx 0.3069$

Total cell-state gradient (both channels combined)

$$\frac{\partial L_2}{\partial C_4^{(2)}} = 0.003042 \times 0.5499 + 0.009167 \approx 0.010840$$

Activation derivatives at the pre-activations

- $\sigma'(\text{net}_{f_4} = 0.0586) \approx 0.2498$
- $\sigma'(\text{net}_{i_4} = 0.2333) \approx 0.2466$
- $\sigma'(\text{net}_{o_4} = 0.3052) \approx 0.2443$
- $\tanh'(\text{net}_{\tilde{C}_4} = 0.2177) \approx 0.9540$

Pre-activation gradients (the four deltas)

- $\delta_{o_4} = 0.003042 \times 0.2116 \times 0.2443 \approx 0.000157$
- $\delta_{f_4} = 0.010840 \times 0.1850 \times 0.2498 \approx 0.000501$
- $\delta_{i_4} = 0.010840 \times 0.2144 \times 0.2466 \approx 0.000573$
- $\delta_{\tilde{C}_4} = 0.010840 \times 0.5580 \times 0.9540 \approx 0.005771$

Output Gate Gradients

- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} (\text{from } t = 4) = 0.000157 \times 0.1043 \approx 0.000016$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} (\text{from } t = 4) = 0.000157 \times 0.2184 \approx 0.000034$
- $\frac{\partial L_2}{\partial b_o^{(2)}} (\text{from } t = 4) = 0.000157 \times 1 \approx 0.000157$

Forget Gate Gradients

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \text{ (from } t = 4) = 0.000501 \times 0.1043 \approx 0.000052$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \text{ (from } t = 4) = 0.000501 \times 0.2184 \approx 0.000109$
- $\frac{\partial L_2}{\partial b_f^{(2)}} \text{ (from } t = 4) = 0.000501 \times 1 \approx 0.000501$

Input-Gate Sigmoid Gradients

- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \text{ (from } t = 4) = 0.000573 \times 0.1043 \approx 0.000060$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \text{ (from } t = 4) = 0.000573 \times 0.2184 \approx 0.000125$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \text{ (from } t = 4) = 0.000573 \times 1 \approx 0.000573$

Candidate Gradients

- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \text{ (from } t = 4) = 0.005771 \times 0.1043 \approx 0.000602$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \text{ (from } t = 4) = 0.005771 \times 0.2184 \approx 0.001260$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \text{ (from } t = 4) = 0.005771 \times 1 \approx 0.005771$

Recurrent Gradients at $t = 4$ (to pass to $t = 3$)

$$\frac{\partial L_2}{\partial h_3^{(2)}} = 0.000157 \times 0.41 + 0.000501 \times 0.33 + 0.000573 \times 0.29 + 0.005771 \times 0.18 \approx 0.001435$$

$$\frac{\partial L_2}{\partial C_3^{(2)}} = 0.010840 \times 0.5146 \approx 0.005579$$

Gradient passed down to Unit 1 at $t = 4$ (through the input connection $h_4^{(1)}$):

$$\frac{\partial L_2}{\partial h_4^{(1)}} = 0.000157 \times 0.24 + 0.000501 \times (-0.21) + 0.000573 \times 0.38 + 0.005771 \times 0.27 \approx 0.001709$$

Time Step $t = 3$

Using Unit 2 forward-pass values at $t = 3$: $h_3^{(2)} \approx 0.1043$, $C_3^{(2)} \approx 0.1850$, $o_3^{(2)} \approx 0.5705$, $f_3^{(2)} \approx 0.5157$, $i_3^{(2)} \approx 0.5515$, $\tilde{C}_3^{(2)} \approx 0.1969$, $h_2^{(2)} \approx 0.0833$, $C_2^{(2)} \approx 0.1481$, input $h_3^{(1)} \approx 0.1649$

Incoming gradients (from the next time step)

- $\frac{\partial L_2}{\partial h_3^{(2)}} \approx 0.001435$
- $\frac{\partial L_2}{\partial C_3^{(2)}} \Big|_{\text{incoming}} \approx 0.005579$

Common building blocks

- $\tanh(C_3^{(2)}) = \tanh(0.1850) \approx 0.1829, \quad 1 - \tanh^2(C_3^{(2)}) \approx 0.9665$
- $\frac{\partial h_3}{\partial o_3} = \tanh(C_3^{(2)}) \approx 0.1829$
- $\frac{\partial h_3}{\partial C_3} = 0.5705 \odot 0.9665 \approx 0.5514$
- $\frac{\partial h_3}{\partial f_3} = 0.5514 \odot 0.1481 \approx 0.0817$
- $\frac{\partial h_3}{\partial i_3} = 0.5514 \odot 0.1969 \approx 0.1086$
- $\frac{\partial h_3}{\partial \tilde{C}_3} = 0.5514 \odot 0.5515 \approx 0.3041$

Total cell-state gradient (both channels combined)

$$\frac{\partial L_2}{\partial C_3^{(2)}} = 0.001435 \times 0.5514 + 0.005579 \approx 0.006370$$

Activation derivatives at the pre-activations

- $\sigma'(\text{net}_{f_3} = 0.0629) \approx 0.2498$
- $\sigma'(\text{net}_{i_3} = 0.2068) \approx 0.2473$
- $\sigma'(\text{net}_{o_3} = 0.2837) \approx 0.2450$
- $\tanh'(\text{net}_{\tilde{C}_3} = 0.1995) \approx 0.9612$

Pre-activation gradients (the four deltas)

- $\delta_{o_3} = 0.001435 \times 0.1829 \times 0.2450 \approx 0.000064$
- $\delta_{f_3} = 0.006370 \times 0.1481 \times 0.2498 \approx 0.000236$
- $\delta_{i_3} = 0.006370 \times 0.1969 \times 0.2473 \approx 0.000310$
- $\delta_{\tilde{C}_3} = 0.006370 \times 0.5515 \times 0.9612 \approx 0.003377$

Output Gate Gradients

- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} (\text{from } t = 3) = 0.000064 \times 0.0833 \approx 0.000005$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} (\text{from } t = 3) = 0.000064 \times 0.1649 \approx 0.000011$
- $\frac{\partial L_2}{\partial b_o^{(2)}} (\text{from } t = 3) = 0.000064 \times 1 \approx 0.000064$

Forget Gate Gradients

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \text{ (from } t = 3) = 0.000236 \times 0.0833 \approx 0.000020$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \text{ (from } t = 3) = 0.000236 \times 0.1649 \approx 0.000039$
- $\frac{\partial L_2}{\partial b_f^{(2)}} \text{ (from } t = 3) = 0.000236 \times 1 \approx 0.000236$

Input-Gate Sigmoid Gradients

- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \text{ (from } t = 3) = 0.000310 \times 0.0833 \approx 0.000026$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \text{ (from } t = 3) = 0.000310 \times 0.1649 \approx 0.000051$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \text{ (from } t = 3) = 0.000310 \times 1 \approx 0.000310$

Candidate Gradients

- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \text{ (from } t = 3) = 0.003377 \times 0.0833 \approx 0.000281$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \text{ (from } t = 3) = 0.003377 \times 0.1649 \approx 0.000557$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \text{ (from } t = 3) = 0.003377 \times 1 \approx 0.003377$

Recurrent Gradients at $t = 3$ (to pass to $t = 2$)

$$\frac{\partial L_2}{\partial h_2^{(2)}} = 0.000064 \times 0.41 + 0.000236 \times 0.33 + 0.000310 \times 0.29 + 0.003377 \times 0.18 \approx 0.000802$$

$$\frac{\partial L_2}{\partial C_2^{(2)}} = 0.006370 \times 0.5157 \approx 0.003285$$

Gradient passed down to Unit 1 at $t = 3$ (through the input connection $h_3^{(1)}$):

$$\frac{\partial L_2}{\partial h_3^{(1)}} = 0.000064 \times 0.24 + 0.000236 \times (-0.21) + 0.000310 \times 0.38 + 0.003377 \times 0.27 \approx 0.000996$$

Time Step $t = 2$

Using Unit 2 forward-pass values at $t = 2$: $h_2^{(2)} \approx 0.0833$, $C_2^{(2)} \approx 0.1481$, $o_2^{(2)} \approx 0.5665$, $f_2^{(2)} \approx 0.5132$, $i_2^{(2)} \approx 0.5483$, $\tilde{C}_2^{(2)} \approx 0.1890$, $h_1^{(2)} \approx 0.0482$, $C_1^{(2)} \approx 0.0867$, input $h_2^{(1)} \approx 0.1578$

Incoming gradients (from the next time step)

- $\frac{\partial L_2}{\partial h_2^{(2)}} \approx 0.000802$
- $\frac{\partial L_2}{\partial C_2^{(2)}} \Big|_{\text{incoming}} \approx 0.003285$

Common building blocks

- $\tanh(C_2^{(2)}) = \tanh(0.1481) \approx 0.1471, \quad 1 - \tanh^2(C_2^{(2)}) \approx 0.9784$
- $\frac{\partial h_2}{\partial o_2} = \tanh(C_2^{(2)}) \approx 0.1471$
- $\frac{\partial h_2}{\partial C_2} = 0.5665 \odot 0.9784 \approx 0.5543$
- $\frac{\partial h_2}{\partial f_2} = 0.5543 \odot 0.0867 \approx 0.0481$
- $\frac{\partial h_2}{\partial i_2} = 0.5543 \odot 0.1890 \approx 0.1047$
- $\frac{\partial h_2}{\partial \tilde{C}_2} = 0.5543 \odot 0.5483 \approx 0.3039$

Total cell-state gradient (both channels combined)

$$\frac{\partial L_2}{\partial C_2^{(2)}} = 0.000802 \times 0.5543 + 0.003285 \approx 0.003729$$

Activation derivatives at the pre-activations

- $\sigma'(\text{net}_{f_2} = 0.0528) \approx 0.2498$
- $\sigma'(\text{net}_{i_2} = 0.1939) \approx 0.2477$
- $\sigma'(\text{net}_{o_2} = 0.2676) \approx 0.2456$
- $\tanh'(\text{net}_{\tilde{C}_2} = 0.1913) \approx 0.9643$

Pre-activation gradients (the four deltas)

- $\delta_{o_2} = 0.000802 \times 0.1471 \times 0.2456 \approx 0.000029$
- $\delta_{f_2} = 0.003729 \times 0.0867 \times 0.2498 \approx 0.000081$
- $\delta_{i_2} = 0.003729 \times 0.1890 \times 0.2477 \approx 0.000175$
- $\delta_{\tilde{C}_2} = 0.003729 \times 0.5483 \times 0.9643 \approx 0.001972$

Output Gate Gradients

- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} (\text{from } t = 2) = 0.000029 \times 0.0482 \approx 0.000001$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} (\text{from } t = 2) = 0.000029 \times 0.1578 \approx 0.000005$
- $\frac{\partial L_2}{\partial b_o^{(2)}} (\text{from } t = 2) = 0.000029 \times 1 \approx 0.000029$

Forget Gate Gradients

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \text{ (from } t = 2) = 0.000081 \times 0.0482 \approx 0.000004$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \text{ (from } t = 2) = 0.000081 \times 0.1578 \approx 0.000013$
- $\frac{\partial L_2}{\partial b_f^{(2)}} \text{ (from } t = 2) = 0.000081 \times 1 \approx 0.000081$

Input-Gate Sigmoid Gradients

- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \text{ (from } t = 2) = 0.000175 \times 0.0482 \approx 0.000008$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \text{ (from } t = 2) = 0.000175 \times 0.1578 \approx 0.000028$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \text{ (from } t = 2) = 0.000175 \times 1 \approx 0.000175$

Candidate Gradients

- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \text{ (from } t = 2) = 0.001972 \times 0.0482 \approx 0.000095$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \text{ (from } t = 2) = 0.001972 \times 0.1578 \approx 0.000311$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \text{ (from } t = 2) = 0.001972 \times 1 \approx 0.001972$

Recurrent Gradients at $t = 2$ (to pass to $t = 1$)

$$\frac{\partial L_2}{\partial h_1^{(2)}} = 0.000029 \times 0.41 + 0.000081 \times 0.33 + 0.000175 \times 0.29 + 0.001972 \times 0.18 \approx 0.000444$$

$$\frac{\partial L_2}{\partial C_1^{(2)}} = 0.003729 \times 0.5132 \approx 0.001914$$

Gradient passed down to Unit 1 at $t = 2$ (through the input connection $h_2^{(1)}$):

$$\frac{\partial L_2}{\partial h_2^{(1)}} = 0.000029 \times 0.24 + 0.000081 \times (-0.21) + 0.000175 \times 0.38 + 0.001972 \times 0.27 \approx 0.000589$$

Time Step $t = 1$

Using Unit 2 forward-pass values at $t = 1$: $h_1^{(2)} \approx 0.0482$, $C_1^{(2)} \approx 0.0867$, $o_1^{(2)} \approx 0.5573$, $f_1^{(2)} \approx 0.5131$, $i_1^{(2)} \approx 0.5379$, $\tilde{C}_1^{(2)} \approx 0.1612$, $h_0^{(2)} \approx 0.0000$, $C_0^{(2)} \approx 0.0000$, input $h_1^{(1)} \approx 0.0838$

Incoming gradients (from the next time step)

- $\frac{\partial L_2}{\partial h_1^{(2)}} \approx 0.000444$
- $\frac{\partial L_2}{\partial C_1^{(2)}} \Big|_{\text{incoming}} \approx 0.001914$

Common building blocks

- $\tanh(C_1^{(2)}) = \tanh(0.0867) \approx 0.0865, \quad 1 - \tanh^2(C_1^{(2)}) \approx 0.9925$
- $\frac{\partial h_1}{\partial o_1} = \tanh(C_1^{(2)}) \approx 0.0865$
- $\frac{\partial h_1}{\partial C_1} = 0.5573 \odot 0.9925 \approx 0.5531$
- $\frac{\partial h_1}{\partial f_1} = 0.5531 \odot 0.0000 \approx 0.0000$
- $\frac{\partial h_1}{\partial i_1} = 0.5531 \odot 0.1612 \approx 0.0892$
- $\frac{\partial h_1}{\partial \tilde{C}_1} = 0.5531 \odot 0.5379 \approx 0.2975$

Total cell-state gradient (both channels combined)

$$\frac{\partial L_2}{\partial C_1^{(2)}} = 0.000444 \times 0.5531 + 0.001914 \approx 0.002159$$

Activation derivatives at the pre-activations

- $\sigma'(\text{net}_{f_1} = 0.0524) \approx 0.2498$
- $\sigma'(\text{net}_{i_1} = 0.1518) \approx 0.2486$
- $\sigma'(\text{net}_{o_1} = 0.2301) \approx 0.2467$
- $\tanh'(\text{net}_{\tilde{C}_1} = 0.1626) \approx 0.9740$

Pre-activation gradients (the four deltas)

- $\delta_{o_1} = 0.000444 \times 0.0865 \times 0.2467 \approx 0.000009$
- $\delta_{f_1} = 0.002159 \times 0.0000 \times 0.2498 \approx 0.000000$
- $\delta_{i_1} = 0.002159 \times 0.1612 \times 0.2486 \approx 0.000087$
- $\delta_{\tilde{C}_1} = 0.002159 \times 0.5379 \times 0.9740 \approx 0.001131$

Output Gate Gradients

- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} (\text{from } t = 1) = 0.000009 \times 0.0000 \approx 0.000000$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} (\text{from } t = 1) = 0.000009 \times 0.0838 \approx 0.000001$
- $\frac{\partial L_2}{\partial b_o^{(2)}} (\text{from } t = 1) = 0.000009 \times 1 \approx 0.000009$

Forget Gate Gradients

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \text{ (from } t = 1) = 0.000000 \times 0.0000 \approx 0.000000$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \text{ (from } t = 1) = 0.000000 \times 0.0838 \approx 0.000000$
- $\frac{\partial L_2}{\partial b_f^{(2)}} \text{ (from } t = 1) = 0.000000 \times 1 \approx 0.000000$

Input-Gate Sigmoid Gradients

- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \text{ (from } t = 1) = 0.000087 \times 0.0000 \approx 0.000000$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \text{ (from } t = 1) = 0.000087 \times 0.0838 \approx 0.000007$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \text{ (from } t = 1) = 0.000087 \times 1 \approx 0.000087$

Candidate Gradients

- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \text{ (from } t = 1) = 0.001131 \times 0.0000 \approx 0.000000$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \text{ (from } t = 1) = 0.001131 \times 0.0838 \approx 0.000095$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \text{ (from } t = 1) = 0.001131 \times 1 \approx 0.001131$

Recurrent Gradients at $t = 1$ (to pass to $t = 0$)

$$\frac{\partial L_2}{\partial h_0^{(2)}} = 0.000009 \times 0.41 + 0.000000 \times 0.33 + 0.000087 \times 0.29 + 0.001131 \times 0.18 \approx 0.000233$$

$$\frac{\partial L_2}{\partial C_0^{(2)}} = 0.002159 \times 0.5131 \approx 0.001108$$

Gradient passed down to Unit 1 at $t = 1$ (through the input connection $h_1^{(1)}$):

$$\frac{\partial L_2}{\partial h_1^{(1)}} = 0.000009 \times 0.24 + 0.000000 \times (-0.21) + 0.000087 \times 0.38 + 0.001131 \times 0.27 \approx 0.000341$$

Final Accumulated Gradients for Unit 2 (after processing $t = 5 \rightarrow t = 1$)

- $\frac{\partial L_2}{\partial w_{hf}^{(2)}} \approx 0.000192$
- $\frac{\partial L_2}{\partial w_{xf}^{(2)}} \approx 0.000375$

- $\frac{\partial L_2}{\partial b_f^{(2)}} \approx 0.001771$
- $\frac{\partial L_2}{\partial w_{hi1}^{(2)}} \approx 0.000211$
- $\frac{\partial L_2}{\partial w_{xi1}^{(2)}} \approx 0.000427$
- $\frac{\partial L_2}{\partial b_{i1}^{(2)}} \approx 0.002103$
- $\frac{\partial L_2}{\partial w_{hi2}^{(2)}} \approx 0.002133$
- $\frac{\partial L_2}{\partial w_{xi2}^{(2)}} \approx 0.004351$
- $\frac{\partial L_2}{\partial b_{i2}^{(2)}} \approx 0.021725$
- $\frac{\partial L_2}{\partial w_{ho}^{(2)}} \approx 0.000244$
- $\frac{\partial L_2}{\partial w_{xo}^{(2)}} \approx 0.000458$
- $\frac{\partial L_2}{\partial b_o^{(2)}} \approx 0.002076$

Weight Update for Unit 2 (Gradient Descent, Single Epoch)

Using learning rate $\alpha = 0.1$:

$$w_{\text{new}} = w_{\text{old}} - \alpha \times \frac{\partial L_2}{\partial w}$$

All 12 updates:

- $w_{hf}^{(2)} \leftarrow 0.33 - 0.1 \times (0.000192) \approx 0.329981$
- $w_{xf}^{(2)} \leftarrow -0.21 - 0.1 \times (0.000375) \approx -0.210038$
- $b_f^{(2)} \leftarrow 0.07 - 0.1 \times (0.001771) \approx 0.069823$
- $w_{hi1}^{(2)} \leftarrow 0.29 - 0.1 \times (0.000211) \approx 0.289979$
- $w_{xi1}^{(2)} \leftarrow 0.38 - 0.1 \times (0.000427) \approx 0.379957$
- $b_{i1}^{(2)} \leftarrow 0.12 - 0.1 \times (0.002103) \approx 0.119790$
- $w_{hi2}^{(2)} \leftarrow 0.18 - 0.1 \times (0.002133) \approx 0.179787$
- $w_{xi2}^{(2)} \leftarrow 0.27 - 0.1 \times (0.004351) \approx 0.269565$

- $b_{i2}^{(2)} \leftarrow 0.14 - 0.1 \times (0.021725) \approx 0.137828$
- $w_{ho}^{(2)} \leftarrow 0.41 - 0.1 \times (0.000244) \approx 0.409976$
- $w_{xo}^{(2)} \leftarrow 0.24 - 0.1 \times (0.000458) \approx 0.239954$
- $b_o^{(2)} \leftarrow 0.21 - 0.1 \times (0.002076) \approx 0.209792$

Note on the Gradient Flowing Down into Unit 1

Because the units are stacked, the loss L_2 also depends on Unit 1's weights through the input connections $h_t^{(1)}$. The gradient sent down to Unit 1 at each time step, $\frac{\partial L_2}{\partial h_t^{(1)}}$, was computed above for all five time steps:

$$\frac{\partial L_2}{\partial h_1^{(1)}} \approx 0.000341, \quad \frac{\partial L_2}{\partial h_2^{(1)}} \approx 0.000589, \quad \frac{\partial L_2}{\partial h_3^{(1)}} \approx 0.000996, \quad \frac{\partial L_2}{\partial h_4^{(1)}} \approx 0.001709, \quad \frac{\partial L_2}{\partial h_5^{(1)}} \approx 0.003158$$

In a full end-to-end training pass of the stacked network, these values would be injected into Unit 1's backward pass (added to Unit 1's incoming $\frac{\partial L}{\partial h_t}$ at each time step) so that Unit 1's weights also learn from L_2 . Part I has been left exactly as in the original document, which trains Unit 1 against its own target; extending Unit 1's pass to include these injected gradients uses the identical five-step procedure listed at the end of Part I.

This completes the full backpropagation of the second LSTM unit. Every incoming gradient, building block, activation derivative, delta, weight gradient, and recurrent gradient has been shown with explicit arithmetic for all 5 time steps, followed by the accumulated gradients and the single-epoch weight update for all 12 of Unit 2's weights.